Energy in Collisions – Elastic and Inelastic

Synopsis: When two bodies collide, one of two things will happen: they will either collide and separate, or they will stick together and move as a unit. When they bounce, it's called an elastic collision. When they stick, it's called an inelastic collision. Note 8 showed that momentum is conserved in both — the sum before equals the sum after, whatever the carts do. This Note asks the other question: what happens to the kinetic energy?

The answer is the real difference between the two kinds. In a perfectly elastic collision the total kinetic energy after is the same as before — the carts' bumpers store it as elastic PE for an instant and give all of it back. In an inelastic collision some of the kinetic energy is gone, into thermal energy, sound, and the deformation of whatever stuck; in a perfectly inelastic collision (they stick) as much is lost as the momentum law allows. Momentum never has a choice. Kinetic energy always does.

Momentum and kinetic energy also answer different questions about stopping something. Both are built from the same m and v, but $p = mv$ and $KE = \frac{1}{2}mv^2$ weigh the velocity differently, and a bullet and a freight car make the difference impossible to miss.

Performance Expectations: Distinguish elastic from inelastic collisions by what happens to the kinetic energy. Calculate the kinetic energy before and after a collision and the fraction lost. Explain why momentum and kinetic energy give different answers about stopping a moving object.

For PHY 151 — with algebra: Derive the relative-velocity rule for a one-dimensional elastic collision, $v_2' - v_1' = v_1 - v_2$, from the two conservation laws.

Instructions and Questions

1) Two kinds of collision, told apart by the kinetic energy.
   In the smart-cart lab (April 2025) cart A, 0.512 kg, was sent into cart B, 0.514 kg, at rest. A stopped dead and B moved off at A's speed. Take $v_A = 0.40$ m/s.
   i) Write the total momentum before and after. Is it conserved?
   ii) Write the total kinetic energy before and after. Is it conserved? What kind of collision is this?
   iii) Two equal carts swapped velocities. Say why that is the only elastic outcome for equal masses with one at rest — what would be wrong with both carts moving off at half speed?

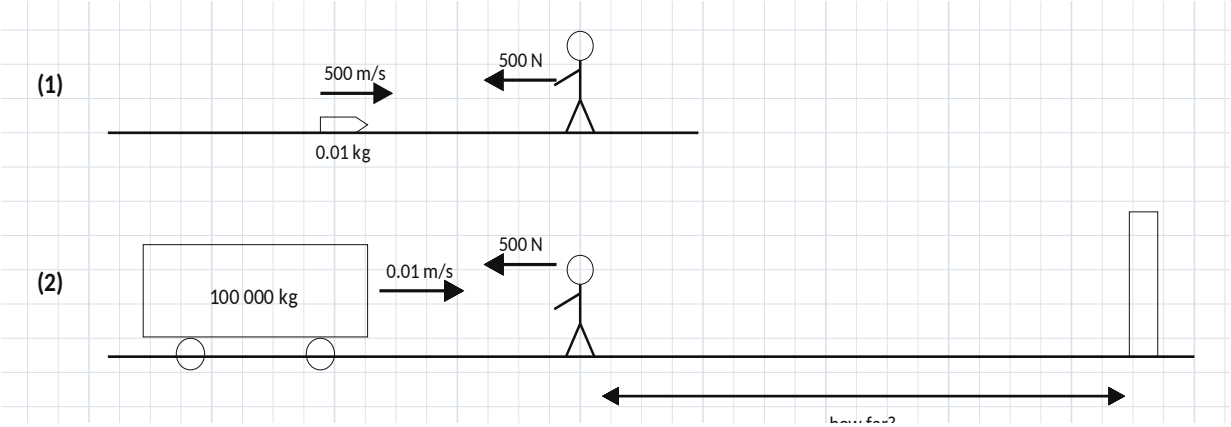
Momentum	Kinetic energy

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2) Perfectly inelastic: as much as the law allows.
   From the same lab: cart A, 0.51 kg, at 0.75 m/s into a loaded cart B, 1.5 kg, at rest. They lock together.
   i) Predict the speed of the pair from momentum conservation.
   ii) Calculate the kinetic energy before and after, and the fraction that was lost.
   iii) The lost kinetic energy did not vanish. Say where it went, and why the momentum did not go with it.
   iv) Show that the fraction lost is $m_B/(m_A + m_B)$ — so a heavy target swallows more of the energy.

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3) Same person, same push: a bullet and a freight car.



A 0.01 kg bullet at 500 m/s, and a 100 000 kg freight car at 0.01 m/s. In each case a person pushes back with 500 N until the thing stops.

i) Calculate the momentum and the kinetic energy of each.
ii) Use the work–energy relation (Note 4) to find how far each one travels against the 500 N before stopping.
iii) Use the impulse–momentum relation (Note 9) to find how long each one takes to stop.
iv) In two sentences: which quantity sets the stopping distance, which sets the stopping time, and why you cannot stop the car with your hands even though it is barely moving.

The relative-velocity rule.

For a one-dimensional elastic collision the two conservation laws are $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$ (momentum) and $\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}m_1v_1'^2 + \frac{1}{2}m_2v_2'^2$ (kinetic energy). Move the m_1 terms to the left

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and the m_2 terms to the right in both, factor the energy equation with $a^2 - b^2 = (a + b)(a - b)$, and divide it by the momentum equation.

i) Carry the derivation through to $v_2' - v_1' = v_1 - v_2$: the relative velocity reverses.

ii) Use the rule with momentum conservation to show that equal masses with one at rest must swap velocities.

iii) Two carts of masses m_1 and m_2 approach head-on at equal speeds v. Show that $(m_1 + m_2)v_1' = (m_1 - 3m_2)v$, and say what happens to cart 1 when $m_1 = 3m_2$.

4) Looking ahead.

That is the module: two conserved quantities, one that is always kept and one that is kept only when nothing is lost to heat. The two in-class activities that follow put both to work on the carts.

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